cost of living index number measures the general price movement of the commodities consumed by different classes of people.

Uses of Cost of Living Index Number

(i) It indicates whether the real wages of workers are rising or falling for a given time.

(ii) It is used by the administrators for regulating dearness allowance or grant of bonus to the workers.

Methods of constructing Cost of Living Index Number

The cost of living index number can be constructed by the following methods,

(i) Aggregate Expenditure Method (or) Weighted Aggregative Method.

(ii) Family Budget Method.

Aggregate Expenditure Method

This is the most common method used to calculate cost of living index number. In this method, weights are assigned to various commodities consumed by a group in the base year. In this method the quantity of the base year is used as weight.

The formula is given by,

$$\text{Cost of Living Index Number} = \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100$$

Note

The formula for Aggregate Expenditure Method to calculate Cost of Living Index Number is same as formula of Laspeyre's Method.

Family Budget Method

In this method, the weights are calculated by multiplying prices and quantity of the base year. (i.e.) $V = \sum p_0 q_0$. The formula is given by,

$$\text{Cost of Living Index Number} = \frac{\sum PV}{\sum V}$$

where $P = \frac{p_1}{p_0} \times 100$ is the price relative.

$V = \sum p_0 q_0$ is the value relative.

Note

This method is same as the weighted average of price relative method.

When to Use ?

When the Price and Quantity are given, Aggregate Expenditure Method is used.

When the Price and Weight are given, Family Budget Method is used.

Example 9.15

Calculate the cost of living index number for the following data.

Commodities	Quantity 2005	Price	
		2005	2010
A	10	7	9
B	12	6	8
C	17	10	15
D	19	14	16
E	15	12	17

Solution:

Commodities	Quantity 2005 (Q_0)	Price		P_1Q_0	P_0Q_0
		2005 (P_0)	2010 (P_1)		
A	10	7	9	90	70
B	12	6	8	96	72
C	17	10	15	255	170
D	19	14	16	304	266
E	15	12	17	255	180
			Total	1000	758

Table 9.14

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